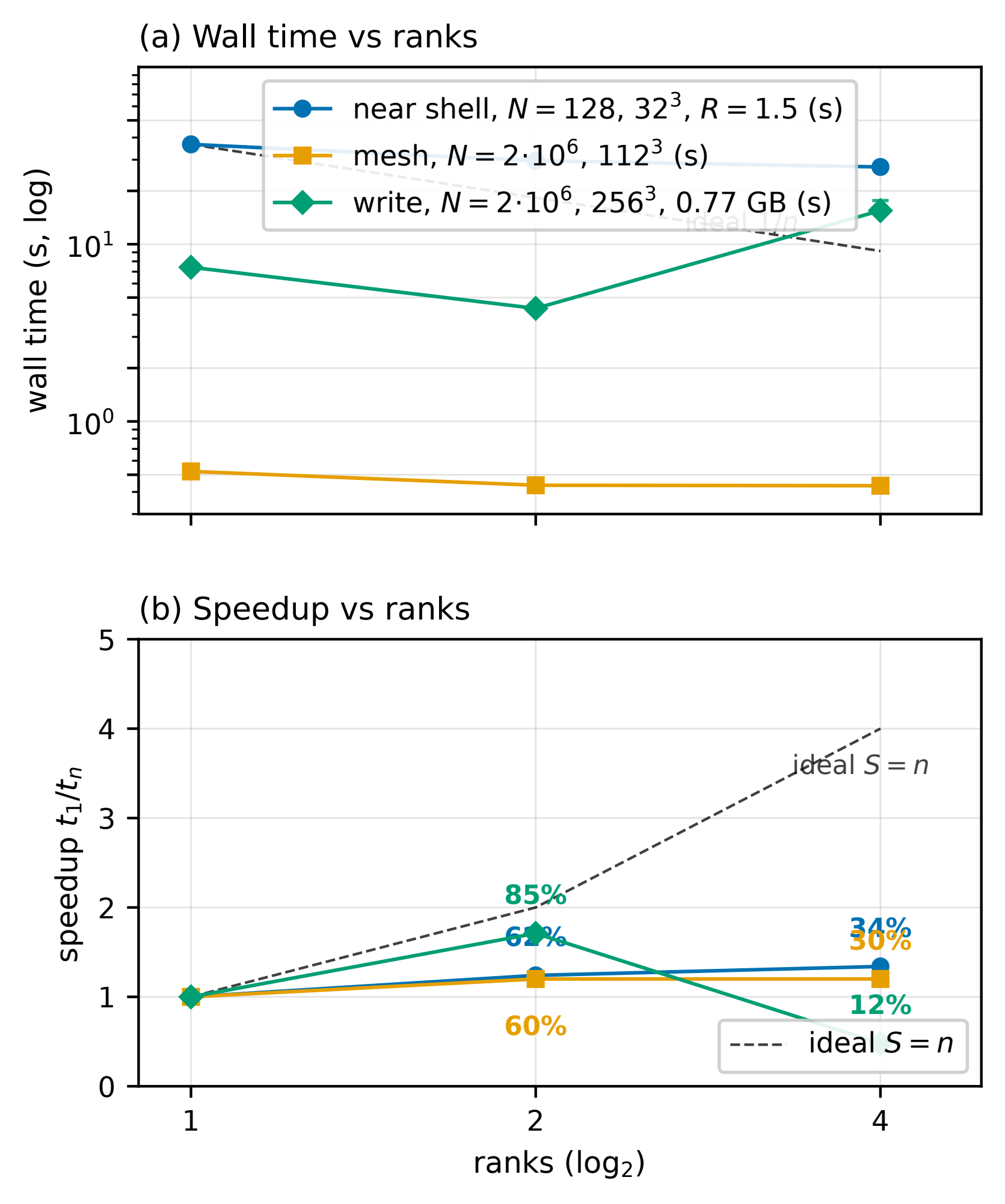
(a) Wall time vs ranks
wall time (s, log)
10^1
10^0
near shell, $N = 128$, 32^3, $R = 1.5$ (s)
mesh, $N = 2 \cdot 10^6$, 112^3 (s)
write, $N = 2 \cdot 10^6$, 256^3, 0.77 GB (s)
ideal $1/n$
(b) Speedup vs ranks
speedup t_1/t_n
0
1
2
3
4
5
ideal $S = n$
85%
34%
30%
12%
60%
ideal $S = n$
1
2
4
ranks ($\log_2$)